

MODULE 1. GENERAL FIRST AID

DEFINITION OF FIRST AID

First Aid is an immediate care given to a person who has been injured or suddenly taken ill. It includes selfhelp and home care if medical assistance is not available or delayed.

ROLES OF FIRST AID

1. It is the bridge that fills the gap between the victim and the physician.
2. It is not intended to compete with, nor take the place of the services of the physician.
3. It ends when the services of a physician begins.

OBJECTIVES OF FIRST AID

1. To alleviate suffering
2. To prevent added/further injury or danger
3. To prolong life

NEED AND VALUE OF FIRST AID

1. To minimize if not totally prevent accident.
2. To prevent added injury or danger.
3. To train people to do the right thing at the right time.
4. Accident happens and sudden illnesses are common and often serious.
5. People very often harm rather than help.
6. Proper and immediate care is necessary to save life or limb.

GUIDELINES FOR GIVING EMERGENCY CARE

I. **Getting started**

- I.1. Planning of action
- I.2. Gathering of needed materials
- I.3. Initial response as follows:

- A Ask for help
- I Intervene
- D Do not further harm

Ask for help. In a crisis, time is of essence. The more quickly you recognize an emergency, and the faster you call for medical assistance, the sooner the victim will get help. Immediate care can greatly affect the outcome of an emergency.

Intervene. To intervene means to do something for the victim that will help achieve a positive outcome to an emergency. Sometimes getting medical help will be all you can do, and this alone may save a life. In other situation, however, you may become actively involved in the victim's initial care by giving first aid. Let the golden rules of emergency care guide your effort.

Do no further harm. Once you have begun first aid, you want to be certain you don't do anything that might cause the victim's condition to worsen. Certain actions should always be avoided by keeping them in mind, you will be able to avoid adding to or worsening the victim's illness or injuries.

- I.4. Instruct helpers

2. **"Emergency Action Principles"**

- 2.1. Survey the scene
- 2.2. Do a primary survey of the victim
- 2.3. Activate medical assistance/transfer facility
- 2.4. Do a secondary survey of the victim

Survey the scene

- o is the scene safe?
- o what happened?
- o how many people are injured?
- o are there bystanders who can help?
- o Identify yourself as a trained first aider.

Do a primary survey of the victim

Check for vital body functions: BREATHING and CIRCULATION by following the ABC steps

- A** Airway - *Is the victim conscious?*
- o If the victim is conscious, assess breathing as described in B.
 - o If the victim is unconscious, start immediately airway management (open the airway refer to Module 4).
- B** Breathing - *Is the victim breathing?*
- o If the victim is breathing,
 - is it shallow or deep?
 - does he appear to be choking?
 - is he cyanotic, suggesting poor oxygenation?
 - o If the victim appears to have any difficulty breathing, immediately support his breathing (maintain adequate open airway).
 - o If the victim is not breathing, provide initial ventilation (refer to Module 4).
- C** Circulation - *Is the victim's heart beating?*
- o If it is, then how is it? (assess pulse) provide other care as necessary.
 - o If not, perform CPR refer to Module 5.
 - Is he severely bleeding?
 - o If he is, control bleeding refer to Module 9.

Activate medical assistance (AMA) or Transfer Facility

(In some emergencies, you'll have enough time to call for specific medical advice before administering first aid. But in some situations, you'll need to attend to the victim first.)

Depending on the situation:

- o a bystander should make the telephone call for help (if available).
- o a bystander will be requested to call for a physician.
- o somebody will be asked to arrange for transfer facility.

Information to be remembered in activating medical assistance:

- o what happened
- o number of persons injured
- o extent of injury and first aid given
- o the telephone number from where you are calling
- o person who activated medical assistance must drop the phone last.

Do a secondary survey of the victim

Interview the victim:

- o introduce yourself
- o get permission to give care
- o ask the victim's name
- o ask what happened
- o ask "do you have any pain or discomfort?"
- o "do you have any allergies?"
- o "are you taking any medication?"

Check the vital signs:

- o determine radial or carotid pulse
(pulse rate per minute:)

Adult 60 - 90/min. Child 80 - 100/min. Infant 100 - 120/min.
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- o determine breathing (respiration rate)
- o determine skin appearance
- o look at the victim's face and lips
- o record skin appearance
 - temperature
 - moisture
 - color
- o do the head to toe examination:
 - Start with the head.
 - Look and feel for cut, bruises and other signs of injury.
 - Check and compare pupils of both eyes

... dilated pupils involve bleeding and state of shock ... constricted pupils may mean heat stroke or drug overdose. ... unequal pupils may suspect head injury or stroke.
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- Check for fluid or blood in ears, nose and mouth.
- Gently feel the sides of the neck for signs of injury.
- Check and compare both collar bones and shoulders
- Check the chest and rib cage.
- Check the victim's abdomen for tenderness by pressing lightly with flat part of your fingers.
- Check the hip bone by pressing slowly downward and inward for possible fracture.
- Check one leg at a time.
- Check one arm at a time.
- Check the spinal column by placing the victim into side lying down position and press gently from the cervical region down to the lumbar for possible injury.
- o record all the assessment including the time.
- o keep the injured person lying down, his head level with his feet.
- o keep the injured person warm and guard against chilling.

3. The golden rules of emergency care

3.1. What to do:

- Do* obtain consent, when possible.
- Do* think the worst, it's best to administer first aid for the gravest possibility.
- Do* call or send for help.
- Do* remember to identify yourself to the victim.
- Do* provide comfort and emotional support.
- Do* respect the victim's modesty and physical privacy
- Do* be as calm and as direct as possible
- Do* care for the most serious injuries first.
- Do* assist the victim with his or her prescription medication.
- Do* keep onlookers away from the injured person.
- Do* handle the victim to a minimum.
- Do* loosen tight clothing.

3.2. What not to do:

Do not let the victim see his own injury.

Do not leave the victim alone except to get help.

Do not assume that the victim's obvious injuries are the only ones.

Do not deny a victim's physical or emotional coping limitation.

Do not further harm the victim like the following:

o trying to arouse an unconscious victim.

o administering fluid/alcoholic drink.

Do not make any unrealistic promises.

Do not trust the judgement of a confused victim.

Do not require the victim to make decisions.

CHARACTERISTICS OF A GOOD FIRST AIDER:

1. Observant - should notice all signs.
2. Resourceful - should make the best use of things at hand
3. Gentle - should not cause pain
4. Tactful - should not alarm the victim
5. Sympathetic - should be comforting

CLOTH MATERIALS COMMONLY USED IN FIRST AID

I. Dressing or Compress

I.1. Definition: any sterile cloth materials used to cover the wound

I.2. Other uses of a dressing or compress:

.2.1. control bleeding

.2.2. protects the wound from infection

.2.3. absorbs liquid from the wound such as blood plasma, water and pus.

I.3. Kinds of dressing:

.3.1. roller gauze

.3.2. square or eye pads

.3.3. compress or adhesive (two types:)

- occlusive dressing

- butterfly dressing

I.4. Application

.4.1. completely cover the wound

.4.2. avoid contamination when handling and applying

2. Bandages

2.1. Definition: any clean cloth materials sterile or not use to hold the dressing in place.

2.2. Other uses of bandage:

.2.1. control bleeding

.2.2. tie splints in place

.2.3. immobilize body part

.2.4. for arm support - use as a sling

2.3. Kinds:

.3.1. triangular .3.5. muslin binder

.3.2. cravat .3.6. elastic bandage

.3.3. roller

.3.4. four-tail

2.4. Application:

.4.1. must proper, neat and correct

.4.2. apply snugly not too loose not too tight

.4.3. always check for tightness caused by later swelling

.4.4. tie ends with a square knot

2.5. Triangular Bandage

.5.1. usually made from a 40-inch square piece of cloth, cut from one corner to the opposite to form a triangle.

- .5.2. can be folded to form cravats (broad cravat, semi-broad cravat or narrow broad).
- 2.6. Square knot - use square knot in the ends of bandage.

.6.1. Rule in tying square knot: right end over left end then left end over right end (vice versa)

- .6.2. Advantages of square knot:
 - easy to tie and untie
 - it has a comfortable flat surface
 - once secured, does not slip nor tightened or loosen.

HINDRANCES IN GIVING EMERGENCY CARE

1. Unfavorable surrounding
 - 1.1. night time
 - 1.2. crowded city streets; churches; shopping mall
 - 1.3. busy highways
 - 1.4. cold or rainy weather
 - 1.5. lack of necessary materials or helpers
2. The presence of crowds

- 2.1. crowds curiously watch, sometimes heckle, sometimes offer incorrect advice.
- 2.2. they may demand haste in transportation or attempt other improper procedures.
- 2.3. a good examination is difficult while a crowd look on.

3. Pressures from victims or relatives
 - 3.1. The victim usually welcomes help, but if he is drunk, he is often hard to examine and handle, and is often misleading in his response.
 - 3.2. The hysteria of relatives or the victim, the evidence of pain, blood and possible early death, exert great pressure on the first aider.
 - 3.3. the first aider may fail to examine carefully and may be persuaded to do what he would know in calm moments to be wrong.

The first aider can meet all these difficulties. Forewarned is forearmed. He should remember that few cases demand haste, or good examination is important and can be done slowly, and he has no other job or appointment as important and so gratifying as saving a life or limb.

MODULE 2. THE HUMAN BODY

(Note: The objective of this module is not to let the participants study the human body but to make them understand the parts and functions so that first aid measures of injuries/illnesses are better understood and appreciated.)

THE LANGUAGE OF TOPOGRAPHIC ANATOMY

The surface of the body has many definite visible features that serve as guidelines or landmarks to structures that lie beneath them. These external features or topography give clues to the general anatomy of the body. A sharp awareness of the superficial landmarks of the body - its topographic anatomy will help the well-trained examiner to evaluate the ill or injured person. Visual inspection of the body is the simplest step in primary and secondary surveys.

All emergency medical personnel must be familiar with the topographic anatomy. The use of proper terms will assure the correct information with least possible confusion. The terms used to describe topographic anatomy are applied to the body when it is in the anatomic position, or the position standing erect, facing the examiner, arms at the side and palms forward. When the terms right and left are used, they refer to the patient's right and left. The principal regions of the body are head, neck, thorax (chest), abdomen, and extremities (arms and legs).

The front surface of the body, facing the examiner is the **anterior surface**. The surface of the patient away from the examiner is the **posterior surface**. An imaginary vertical line drawn from the midforehead through the nose and the *umbilicus* (navel) to the floor is termed the **midline of the body**. This imaginary line divides the body into two halves, which are mirror images of each other. Parts of the body that lie distant from the midline are termed **lateral structures**. Parts of the body that lie closer to the midline are termed **medial structures**. For example we speak of the medial (inner) and lateral (outer) of the knee or the eye. The **superior portion** of the body, or any part, is that portion near the head, while a portion nearer the feet is the **inferior portion**. We also use these terms to describe the relationship of one structure to another.

 For example, the nose is superior to the mouth and inferior to the forehead.

The terms proximal and distal are used to describe the relationship of any two structures on a limb. Proximal describes structures that are closer to the trunk. **Distal** describes structures that are nearer to the free end of the extremities.

 For example, the elbow is distal to the shoulder yet proximal to the wrist and hand.

The human body is made up of millions of cells each specialized to carry out its own particular functions but coordinated with all body cells. All cells require food, water and oxygen and the removal of waste products. To do this the human body must have:

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1. A nervous system to coordinate;
 2. A respiratory system to supply oxygen and remove carbon dioxide from the blood;
 3. A circulatory system to transport oxygen, food and water and remove waste products;
 4. A digestive system to absorb food and eliminate some waste products;
 5. A urinary system to remove waste products;
 6. A reproductive system to propagate species;
 7. A skeletal system to give form to the body, allow bodily movement, provide protection to the vital internal organs, produce red blood cells and serves as a reservoir of calcium, phosphorus and other important body chemicals.

- 8. Skin to control body temperature and appreciate sensation.
- 9. Sense organs (the skin, ears, eyes, nose and tongue) to appreciate touch, pain, and temperature, hearing balance, sight, smell and taste.

Thus, oxygen is obtained from the air which we breathe to the lungs. It then enters the bloodstream and distributed to each cell of the body. Carbon dioxide is formed within the cell and is carried by the blood to the lungs to be expelled during exhalation to the air. The food we eat and the water we take is absorbed from the digestive system into the blood. It is utilized by the cells, and waste products formed enter the blood and:

- go to the **kidneys** to be eliminated in the **urine**,
- are passed into the **lower bowel** to be removed in the **feces**,
- are converted to **carbon dioxide** and lost from the **lungs**.

THE NERVOUS SYSTEM

Controlling all activities of the body is the **nervous system**. It consist of the **brain** and the **spinal cord**, with nerves distributed to all organs and tissues of the body. The brain receives, coordinates and reacts to messages received from internal and external sources but also stores information so that it can react from memory. It is also responsible for the control of movements of voluntary muscles.

Motor Nerves: pass from the brain to the muscles of the body to control movements. *Injury to a motor nerve causes paralysis of the muscle supplied.*

Sensory Nerve: Sense organs are situated in the eye, ear, skin, joints, tongue and nose. Sensory nerves receive information from sense organ of sight, hearing, balance, touch, pain, temperature, taste and smell. Sensory nerves lead from these organs to the brain. *Injury to sensory nerves leads to loss of function of the sense organ.*

Damage may be caused to the nervous system by:

- 1. Injury
- 2. Loss of blood supply
- 3. Toxins

Abnormal function of the brain or spinal cord leads to:

- 1. Unconsciousness
- 2. Paralysis
- 3. Malfunction

RESPIRATORY SYSTEM

I. Parts

I.1. Air Passages:

- .1.1. nose and mouth
- .1.2. pharynx
- .1.3. larynx
- .1.4. trachea
- .1.5. bronchial tubes

I.2. Chest Cage:

- .2.1. lungs
- .2.2. heart
- .2.3. ribs and their supports

I.3. Diaphragm

2. Air Inspired and Expired:

Air we take in contains 21 percent oxygen and a trace of carbon dioxide approximately 0.04 percent. For every breath, our body uses only 5 percent of oxygen we inspire to sustain life and produces 4 percent carbon dioxide waste product. During expiration we give off 4 percent carbon dioxide and 16 percent oxygen.

3. Process of Breathing:

When we breath, about 500 ml (1 pint) of air is taken in (**inspiration**), the diaphragm moves downward and the ribs upward and outward. This increases the volume of the chest. A partial vacuum is created in the chest cavity, the lungs expand and the air is sucked in through the mouth and the nose into the lungs. Normal breathing out (**expiration**) is produced by a relaxation of the chest wall and intercostal muscles and moving up of the diaphragm. This forces air out of the lungs.

The amount of air supplied to the blood is controlled by a center in the brain at the base of the skull and in the upper part of the spinal cord (**respiratory center**). This center controls respiration by analyzing the carbon dioxide content of the blood it receives. Too much carbon dioxide causes the center to respond by increasing the depth and rate of the breathing and vice-versa.

The normal breathing rate for an adult at rest is from 12-18 times per minute, and a higher rate for children and infants at about 18-25 times per minute and if more oxygen is required as in exercise, fever or in conditions which restrict the normal function of the lungs such as pneumonia.

CIRCULATORY SYSTEM

The circulatory system of the body consist of the circulation of the blood through all the extremities of the body, and it involves the heart, blood vessels, blood and lymph.

I. Parts

- 1.1. heart
- 1.2. blood
- 1.3. blood vessels

2. Functions

1.1 HEART

The heart is a hollow muscular organ about the size of a fist, lying between the lungs, behind the breastbone. It slants obliquely downward to the left side of the chest.

Function as an electromuscular pump having a left and a right chamber, each subdivided into a large and small chamber, provided with valves which aid in the correct circulation of the blood.

Heart (Pulse Rate):
Adult - 60 - 90 beats/min.
Child - 90 - 100 beats/min.
Infant- 100 - 120 beats/min.

1.2 BLOOD

The blood is a red, sticky fluid circulating through the blood vessels, has a peculiar, faint odor, salty in taste and it varies in color from bright scarlet to a bluish red.

Blood is composed of:

I. Red blood cells (**RBC**) (Erythrocytes) - transport oxygen to the tissues of the body and carry carbon dioxide from the tissues to the lungs.

2. White blood cells (**WBC**) (Leukocytes) - defend the body against foreign bodies such as bacteria or combat infection.

3. **Plasma** (fluid part) - carry the food to all parts of the body and waste materials to the organ of excretion.

About one-thirteenth of the weight of human body is blood. A loss of one-third of this is usually fatal.

1.3 BLOOD VESSELS

1. **arteries** - carry the blood from the heart to all parts of the body.

2. **veins** - carry blood back to the heart.

3. **capillaries** - small blood vessels at the end of the arteries.

Course of Blood

1. Dark venous blood laden with carbon dioxide and waste matter picked up in its progress through the body's veins, is drawn into the right atrium as the atrium lies momentarily relaxed.

2. When the atrium is filled up, the valve in its flood opens downward and blood pours into the ventricle below.

3. When the ventricle is full, its smooth pumping pressure closes the valve, which bulges out like a parachute. This same pressure simultaneously opens another set of valves (half-moon shape or non-return valve) and forces the blood out of the ventricle into the artery that leads directly to the lungs.

4. In the thin wall network of the lungs, the dark blood is purified by changing its load of carbon dioxide for oxygen from the outer air.

5. Fresh from the lungs, the blood enters the left atrium. When the atrium is full, the valve opens and the ventricle begins to fill.

6. The ventricle contracts, pushing its cupful of blood into the aorta, the huge artery that leads out from the base of the heart.

7. From the aorta, widest river of life, the red blood branches out, ever more slowly, through arteries and tiny capillaries, to every cell in the body.

The heart repeats this process of contracting and relaxing, day after day, year in, year out.

Course of Important Blood Vessels

Demonstrate the following by chart or model:

o A large artery (aorta) leaves the heart arches, dividing into main branches which go to the head, upper extremities and the lower extremities.

o The two arteries going to either side of the head and neck are called the carotids.

o The artery which goes to either shoulder and arm is called the subclavian. It becomes the axillary artery in the armpit, and the brachial artery as it passes down the arm.

o From the heart arches the aorta descends, dividing finally into two branches

crossing the mid-groin and running toward each thigh and leg, where they become known as the femoral.

DIGESTIVE SYSTEM

I. Parts

I.1 mouth	I.5 liver	I.9 stomach
I.2 salivary glands	I.6 gall bladder	1.10 intestine
1.3 pharynx	I.7 pancreas	1.11 anus
I.4 esophagus	I.8 rectum	

2. Functions

The food we eat is being chewed within the **mouth**. Three pairs of **salivary glands** are located under the tongue, on each side of the lower jaw and on each cheek which produce nearly 1.5 liters of saliva daily. The digestive enzyme in the saliva initiates the digestion of starches. It also serve as a binder and as a lubricant. The food and water we swallow pass the throat along the voice box.

A leaf-shaped valve covering the opening of the trachea is initiated so that liquids and solids are move into the esophagus and away from the trachea. The contraction of the muscle in the **esophagus** propel the food through it to the stomach . Liquids will pass with very little assistance.

The **stomach** is located at the upper left quadrant of the abdominal cavity largely protected by the lower ribs. Muscular contraction in the wall of the stomach and gastric juice convert ingested food to a thoroughly mixed semisolid mass. The main function of the stomach is to receive and store in the large quantity and provide for its movement into the small bowel in regular small amounts. Poisoning or any reaction to trauma may paralyze gastric muscular action thus causing prolong retention of food in the stomach. Pepsin, a digestive enzyme, is produced in the stomach to initiate digestion of proteins.

The **pancreas**, a flat, solid organ, lies behind and below the liver and stomach. It contains two kinds of glands. One set of glands secretes nearly 2 liters of pancreatic juice daily. This juice contains many enzymes that help in the digestion of fat, starch and protein. It flows directly to the intestine through the pancreatic ducts. The other kind of gland called the Islet of Langerhans secretes its products into the blood stream across the capillaries. These islet produce a hormone that regulates the amount of sugar in the blood. It is known as insulin.

The **liver** is located at the upper right quadrant beneath the diaphragm. It is the largest solid organ in the abdomen and consequently one of the most often injured. It has several functions. Poisonous substances produce by digestion are brought to the liver by the blood and are rendered harmless. It also forms factors necessary for blood clotting and for the production of normal plasma. It also produces between 0.5 to 1 liter of bile to assist in the normal digestion of fat.

The liver is also the principal organ for the storage of sugar for immediate use of the body. It also produces many of the factors that aid in the proper regulation of immune responses.

The liver is connected to the intestine by the ducts. The **gall bladder** is an outpouching of a bile duct that serve as a reservoir for produce in the liver. The presence of food in the intestine triggers the contraction of the gall bladder to empty its content. It usually contains 2-3 ounces of bile. When stone is formed at the gall bladder and pass into the bile duct and causes obstruction, it will produce jaundice.

Intestine. Two kinds of intestine are the small and large. The small intestine is so named because of its diameter in comparison with the large intestine. The small intestine receives food from the stomach wherein secretions from the pancreas and liver are mixed

with food for further digestion. It also produce more enzymes and mucus to aid in the digestion.

Appendix is small tube that opens into the first part of the arge intestine in the right lower quadrant of the abdomen. It is 3 to 4 inches long. It easily becomes obstructed and as a result inflamed and infected. Appendicitis, which is the term for this inflammation, is one of the major causes of severe abdominal distress. The appendix has no major known function.

The **spleen**, a major solid organ, is smaller than the liver. It is found in the left upper quadrant of the abdomen, just beneath the diaphragm. It is not required for life nor it is associated with the functions of the digestive tract. It's major function ies in the normal production and destruction of blood cells. Its function, when removed, can be assumed by the liver and bone marrow.

THE URINARY SYSTEM

1.Parts

- | | |
|-------------|---------------------|
| I.1 kidney | I.3 urinary bladder |
| I.2 ureters | I.4 urethra |

2. The urinary system consist of two kidneys which act as filters to remove waste products from the blood. These products are drained via the ureter into the bladder. The bladder holds urine until it can be conveniently expelled from the body via the urethra.

THE REPRODUCTIVE SYSTEM

I. Parts

- | | |
|----------------------|----------------------|
| I.1 male | I.2 female |
| .1.1 testicles | .2.1 ovary |
| .1.2 vasa deferentia | .2.2 fallopian tubes |
| .1.3 Seminal vessels | .2.3 uterus |
| .1.4 prostate gland | .2.4 vagina |
| .1.5 urethra | |
| .1.6 penis | |

2. Functions

In the male, fluids from the prostate gland and from the seminal vesicles mix during intercourse. During intercourse, special mechanism in the nervous system prevent the passage of urine into the urethra. Only seminal fluids, prostatic fluid and sperm pass from the penis into the vagina during ejaculation.

In the female, the ovaries release a mature egg approximately every 28 days. The egg travel through the fallopian tubes to the uterus to the vagina. The vagina receives the sperm during intercourse, when semen and sperm are deposited in it. The sperm may pass into the uterus and fertilize an egg, causing pregnancy. Should the pregnancy come to completion at the end of nine months, the baby will pass through the vagina and be born.

THE SKELETAL SYSTEM

The skeletal system is the framework of the body. It consist of 206 bones joined to each other loosely or firmly by means of ligaments and muscles. The junction between bones are called joints.

The main bony structure are:

1. the skull
2. the vertebrae

3. the pelvis
4. the ribs
5. the bones of the upper and lower limbs

The Skull is divided into:

1. The **face** and **jaws** which form the framework of the features below the eyes and support the structure of the nose and mouth.
2. The **cranium** which provides rigid protection for the enclosed fragile brain. It is made up of a large number of individual bones firmly united together.

The Vertebrae (spinal column)

The spinal column is made up of thirty-three separate bones vertebrae:

- seven located at the neck (cervical)
- twelve at the chest (thoracic)
- five in the loin (lumbar)
- five in the pelvis (sacral) fixed together to form the sacrum
- four fused together to form the coccyx (tail bone) at the base of the spine.

Between the separate vertebrae, there are discs of elastic tissue called intervertebral disc. These allow some movement between the vertebrae and act also as shock absorbers. Enclosed within the vertebral column is the spinal cord. As the cranium protects the brain, so the vertebral column protects the spinal cord.

The Ribs and Sternum

Extending around the chest from thoracic vertebrae, one pair at each vertebra, are twelve pairs of ribs of which the upper ten pairs are connected with the sternum in front through a bridge of cartilage. The main function is to protect the chest and its contents and to give rigidity to the chest walls.

The Bones of the Upper and Lower Limbs

The upper limb is suspended by muscles and ligaments from the trunk. It is supported by two bones, the shoulder blade (scapula) and the collar bone (clavicle).

The bone of the upper arm is the humerus. The bones of the forearm are the radius and ulna, and then come the small bones of the wrist (carpal bones), the hand (metacarpal) and the fingers (phalanges).

The lower limbs are firmly attached to the trunk through a deep socket on the outer side of each pelvic bone into which the rounded upper end of the thigh bone (femur) fits to form the hip joint. The hip bones (pelvis) are anchored to the sacrum. The pelvis forms a bony protection for the contents of the pelvic cavity. The lower leg has the tibia and the fibula and the small bone of the foot (tarsal) connected to the five metatarsal and phalanges.

The Joints

Between bones are joints where bones come together but at which movement can occur. These movements can vary from almost none as in the skull, to the most freely movable joints, the shoulder joints.

In freely movable joints, the joint surfaces are covered with cartilage, which is smooth and minimizes friction. Also in some joints special pieces of cartilages are found; their function is to make the joints fit more snugly.

Each freely movable joint is surrounded by a double layered capsule, each attached to the margins of the surfaces. The inner (synovial) layer of the capsule produces a lubri-

cating fluids which keeps the joint surfaces moist. The outer layer is made up of strong fibrous tissues, thickened in certain areas to form ligaments.

The Ligaments

The ligaments are placed in such a way to bind the bones firmly together, without restricting the normal range of movement of the particular joint.

The Muscles

Muscles are formed of tissues that allows body movement. There are more than 600 muscles in the human body, generally divided in three types.

1. **Skeletal muscles** are also called striated muscle. It is responsible to all body movement resulting from contraction and relaxation.

2. **Smooth muscles** carry out much of the autonomic work of the body. It is also known as involuntary muscles. It is found in the walls of most of the tubular structures of the body. With its contraction and relaxation, it propels or controls the flow of the contents of these structures along their course. Smooth muscle respond only to primitive stimuli such stretching heat or the need to relieve waste.

3. **Cardiac muscle.** The heart is a large muscle comprise of a pair pumps of equal force - one of the lower and one of higher pressure. The heart must function continuously from birth to death. It is a specially adapted involuntary muscles with a very rich blood supply and its own intrinsic regulatory system. Microscopically, it looks different from both skeletal and smooth muscles. Cardiac muscle can tolerate an interruption of its blood supply for only a few seconds. It requires a continues supply of oxygen and glucose for normal function. Because of its special structure and function, cardiac muscle is placed in a separate category.

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MODULE 3. EMERGENCY RESCUE AND TRANSFER

1. EMERGENCY RESCUE - is a procedure for moving a victim from unsafe place to a place of safety.

2. Indications for Emergency Rescue.

- 2.1. Danger of fire or explosion.
- 2.2. Danger of toxic gases or asphyxia due to lack of oxygen.
- 2.3. Serious traffic hazards.
- 2.4. Risk of drowning.
- 2.5. Danger of electrocution.
- 2.6. Danger of collapsing walls.

3. Methods of Rescue

- 3.1. For immediate rescue without any assistance, drag or pull the victim in the direction of the long axis of his body preferably from the shoulder. If possible, minimize lifting or carrying the injured person before checking for injuries --unless you are sure that there is no major fracture or involvement of his neck or spine.
- 3.2. Most of the one-man drags/carries and other transfer methods can be used as methods of rescue.

4. Objectives of the First Aider -pp2-

When it is necessary to remove a person from a life threatening situation, the objectives for the first aider are:

- 4.1. To ensure an open airway and to administer artificial respiration when it is needed.
- 4.2. To control severe bleeding.
- 4.3. To check for injuries.
- 4.4. To immobilize injured parts before extrication of the victim.
- 4.5. To arrange for transportation.
- 4.6. To avoid subjecting the victim to any unnecessary disturbance.

TRANSFER

1. The first aider may need to initiate transfer of the victim to shelter, home or medical aid. Skill in the use of simple techniques of transfer must be practiced and selection and use of the correct method is necessary. Selection will depend upon the following:

- 1.1. Nature and severity of the injury.
- 1.2. Size of the victim.
- 1.3. Physical capabilities of the first aider.
- 1.4. Number of personnel and equipment available.
- 1.5. Nature of evacuation route.
- 1.6. Distance to be covered.
- 1.7. Sex of the victim (last consideration).

2. Pointers to be Observed During Transfer

- 2.1. Victim's airway must be maintained open.
- 2.2. Hemorrhage is controlled.
- 2.3. Victim is safely maintained in the correct position.
- 2.4. Regular check of the victim's condition is made.
- 2.5. Supporting bandages and dressing remain effectively applied.
- 2.6. The method of transfer is safe, comfortable and as speedy as circumstances permit.
- 2.7. The victim's body is moved as one unit.
- 2.8. The taller first aiders stay at the head side of the victim.
- 2.9. First aiders/bearers must observe ergonomics (proper body position [back maintained straight] in lifting weights) in lifting and during transfer of victim.

3. Methods of Transfer

3.1. One-man assist/carries/drag

- .1.1. assist to walk
- .1.2. carry in arms (cradle)
- .1.3. packstrap carry
- .1.4. piggy back carry
- .1.5. fireman's carry
- .1.6. fireman's drag
- .1.7. blanket drag
- .1.8. shoulder drag
- .1.9. cloth drag
- .1.10 feet drag
- .1.11 inclined drag (head first - passing a stairway)

3.2. Two-man assist/carries

- .2.1. assist to walk
- .2.2. four-hand seat
- .2.3. hands as a litter
- .2.4. chair as a litter
- .2.5. carry by extremities
- .2.6. fireman's carry with assistance

3.3. Three-man carries

- .3.1. bearers along side (for narrow alleys)
- .3.2. hammock carry

3.4. Four/six/eight-man carry

3.5. Blanket (demonstrate the insertion, testing and lifting of blanket)

3.6. Improvised stretcher

- two poles with:
 - o blanket
 - o empty sacks
 - o shirts or coats
 - o triangular bandages

3.7. Commercial stretchers

3.8. Ambulance or rescue van

3.9. Other vehicles

4. Command Used in 3 (and above)- man Carries

- 4.1. Ready to kneel Kneel
- 4.2. Hands over the victim Move
- 4.3. Ready to insert Insert
- 4.4. (Place victim on your knees,) Ready to lift Lift
- 4.5. Ready to stand Stand
- 4.6. Leg/head center (face towards leg or head) Face
 - Face towards head only for the following situations:
 - loading victim to an ambulance
 - going towards an elevated way/area
 - place/area where there is no choice to turn
- 4.7. victim's body press to chest Press (for bearers along side only)
- 4.8. Ready to walk, inner foot first .. Walk
- 4.9. Ready to stop Stop
- 4.10 Face center Face
- 4.11 On your knees and rest Kneel
- 4.12 Ready to unload Unload

5. Reminders

- 5.1. All team members must answer "ready" at every instruction given by the leader.
- 5.2. Always kneel with one knee - the knee towards the head side of the victim.
- 5.3. It is difficult for inexperienced helpers to lift and carry a person gently. They need careful guidance. If there is time, it is wise to rehearse the lifting procedure first using a practice subject.

TRIAGE AND DISASTER MANAGEMENT

1. Disaster - a sudden and serious disruption of life caused by nature or humans that create or threaten to create injuries to a number of persons or properties.

2. Three phases of response to a disaster

- 2.1. Alarm phase which is concerned with the immediate activation of adequate and appropriate resources.
- 2.2. Work phase (or implementation phase) - it is sub-divided into four overlapping steps:
 - .2.1. locate - find or determine where the victim/s is/are—
 - .2.2. access - means of going to the victim/s
 - .2.3. stabilize - life-threatening cases are already given necessary care or victim is already out of danger.
 - .2.4. transport - transfer the victim to medical facility.
- 2.3. Let down phase - after the work is completed, all personnel must recover from the stress of the disaster with Critical Incident Stress Debriefing (CISD).

3. Triage - a process use in sorting patients/victims into categories of priority for care and transport based on the severity of injuries and medical emergencies.

3.1. Highest priority

- o patients requiring immediate care and transport.
- o airway and breathing difficulties
- o exsanguinating hemorrhage
- o open chest or abdominal wounds
- o severe head injuries or head injuries with decreasing level of consciousness
- o major or complicated burns
- o tension pneumothorax
- o pericardial tamponade
- o impending shock-
- o complicating severe medical problems, such as diabetes with complications, cardiac disease, pregnancy

3.2. Intermediate priority - patients whose care/treatment and transportation can be delayed temporarily.

- o burns without complications
- o back injuries with or without spinal injuries
- o major, open or multiple fractures
- o eye injuries
- o stable abdominal injuries

3.3. Delayed or low priority - (the walking wounded)

- patients whose care and transportation can be delayed until last.
- o fracture and sprain
- o laceration
- o soft tissue injuries
- o other lesser injuries

3.4. Lowest priority - patients/victims who are dead or near death.

- o devastating injuries
- o little chance of survival

(If resources are limited, these patients must be ignored to enable these resources to be used on “salvageable” patients.)

The cardinal rule of triage is to do the greatest good for the greatest number.

The START System - The START (simple triage and rapid treatment) system is one method of triage that has proven to be very effective. Patient's evaluation is based on three primary observation (BCM): breathing, circulation and mental status.

Under this system patients are tagged for easy recognition.

1. Priority one (red tag) - immediate care; life threatening.
2. Priority two (yellow tag) - urgent care; can delay transport and treatment up to one hour.
3. Priority three (green tag) - delayed care; can delay transport up to three hours.
4. Priority four (black tag) - no care required; patient is dead.

MODULE 4. SHOCK

Many lives have been lost due to shock, the body's physiological reaction to major physical or emotional insult. A tragic fact is that *many of these deaths were needless* because proper preventive measures can eliminate or lessen the danger of shock.

1. The Nature of Shock

Shock is a word used in medicine to describe many varied and often unrelated abnormal condition that affect both mind and body. The meaning of the term may be clarified by mentioning a few classifications of shock which the first aiders may not have considered.

2. Definition - Shock is a depressed condition of many body functions due to the failure of enough blood to circulate throughout the body following serious injury.

3. Kinds of Shock

- 3.1. Cardiogenic shock
- 3.2. Anaphylactic shock
- 3.3. Hypovolemic shock or Hemorrhagic
- 3.4. Psychogenic shock or Emotional
- 3.5. Neurogenic shock
- 3.6. Metabolic shock
- 3.7. Respiratory shock
- 3.8. Septic shock

4. Basic Causes of Shock

- 4.1. Pump failure - the heart can be damaged by intensive muscular disease or injury, so that it fails to act properly as a pump. It does not generate sufficient energy to move blood through the system.
- 4.2. Relative hypovolemia - the blood vessels constituting the container can dilate so that the blood within them even though it is of normal volume, is insufficient to fill the system and provide efficient perfusion.
- 4.3. Hypovolemia - blood or plasma can be lost so that the volume of the fluid contained within the vascular system is insufficient to perfuse all areas well each minute.

5. Causes

- 5.1. Severe bleeding
- 5.2. Crushing injury
- 5.3. Infection
- 5.4. Heart attack
- 5.5. Perforation of stomach ulcer
- 5.6. Shell bomb and bullet wound
- 5.7. Rupture of tubal pregnancies
- 5.8. Anaphylaxis
- 5.9. Starvation and disease may also cause shock

6. Factors which contribute to shock

- 6.1. Pain
- 6.2. Rough handling
- 6.3. Improper transfer
- 6.4. Continuous bleeding
- 6.5. Exposure to extreme cold or excessive heat
- 6.6. Fatigue

7. Dangers of shock

- 7.1. Lead to death
- 7.2. Predisposes body to infection
- 7.3. Lead to loss of body part

8. Signs and symptoms of shock

8.1. Early stage:

- .1.1. face - pale or cyanotic in color
- .1.2. skin - cold and clammy
- .1.3. breathing - irregular
- .1.4. pulse - rapid and weak
- .1.5. nausea and vomiting
- .1.6. weakness
- .1.7. thirsty

8.2. Late stage:

- .2.1. if the condition deteriorates, victim may become apathetic or relatively unresponsive.
- .2.2. eyes will be sunken with vacant expression.
- .2.3. pupils are dilated.
- .2.4. blood vessels may be congested producing mottled appearances.
- .2.5. blood pressure has very low level.
- .2.6. unconsciousness may occur, body temperature falls.

9. Objectives of First Aid

- 9.1. To improve circulation of the blood.
- 9.2. To ensure an adequate supply of oxygen.
- 9.3. To maintain normal body temperature.

10. First Aid and preventive management for shock

10.1. Proper Position

- .1.1. keep the victim lying down flat.
- .1.2. elevate the lower part of the body a foot or so, if injury is severe from eight to twelve inches high. Observe.
- .1.3. place the victim who is having difficulty in breathing, on his back, with his head and shoulder raised.
- .1.4. head Injury - apply pressure on the injury and keep the victim lying flat. Do not elevate head or lower extremities. When color of the face return to normal, elevate head and shoulder and continue giving care to the injury. In chest injury, raise the head and shoulder slightly.
- .1.5. symptoms of nausea and vomiting or unconsciousness keep the victim lying on one side preferably opposite from his injury except for sucking wound and stroke. The position is known as recovery, coma or lateral position.

10.2. Proper body heat

- .2.1. maintain body temperature and victim must not be perspiring nor chilling.
- .2.2. if the weather is warm, the victim need not to be covered.
- .2.3. if victim is cold, inspite of the weather, a blanket may be placed underneath him and cover the body.

NOTE: Do not give anything by mouth including water. If medical care is delayed and patient is complaining of intense thirst, you may wet his/her lips.

11. Classifications of Shock

11.1. Cardiogenic Shock - the victim is in shock as a result of a heart attack. It is caused by a decreased effectiveness of the heart's pumping action which causes the blood pressure to drop. Chronic lung disease will aggravate cardiogenic shock.

.1.1. Signs and Symptoms:

- .1.1. chest pain
- .1.2. pulse irregular
- .1.3. weakness
- .1.4. blood pressure low
- .1.5. cyanosis lips and underneath the fingers
- .1.6. anxious
- .1.7. occasionally patients who have heart attacks vomit.

.1.2. First Aid (Emergency Care)

- .2.1. Proper position.
- .2.2. Loosen all tight clothing.
- .2.3. Cold compress application / Administer oxygen if necessary.
- .2.4. Reassure and calm the victim.

11.2. Anaphylactic Shock - develops when an individual comes in contact with a foreign protein substance known as allergen to which he has become sensitized.

.2.1. Ways in which Anaphylactic Shock occurs:

- .1.1. Injection
- .1.2. Sting
- .1.3. Ingestion
- .1.4. Inhalation

.2.2. Allergic Reactions

- .2.1. Skin - itching, burning sensation, edema (swelling), cyanosis about the lips

.2.3. Respiratory System

- .3.1. Sneeze or perceive an itch in nasal passage
- .3.2. Tightness in chest
- .3.3. Irritating, dry cough
- .3.4. Dyspnea (difficulty in breathing)

.2.4. Circulatory System

- .4.1. Peripheral vascular system citation
- .4.2. Drop of Blood Pressure
- .4.3. Weak pulse
- .4.4. Pallor and dizziness
- .4.5. Fainting and coma may follow

.2.5. Causes

- .5.1. Restlessness and anxiety may precede all other signs.
- .5.2. A weak and rapid pulse (" Thready" or difficult to breath) occur rapidly.
- .5.3. Cold and wet skin (commonly described as "clammy") reflects a major sympathetic nervous system response.
- .5.4. Profuse sweating is common.
- .5.5. Paleness, and later cyanosis, reflect decreasing oxygen delivery to tissue
- .5.6. Shallow, labored, rapid or possibly irregular or gasping respirations (specially in chest injury which is associated with development of shock) are common ---dull and lusterless eyes with dilated pupils occur as the process develop.
- .5.7. thirst may become intense.
- .5.8. nausea and vomiting.
- .5.9. dropping of blood pressure (commonly late stage)
- .5.10. lost of consciousness may occur.

2.6. First Aid

- .6.1. maintain open airway (application of rescue breathing, if needed).
- .6.2. control on obvious external bleeding by direct pressure.
- .6.3. elevate the lower extremities about 8 to 12 inches.
- .6.4. prevent the loss of body heat (do not, however, overload the victim with cover or attempt to warm the body unduly).
- .6.5. splint fracture: splinting will lessen bleeding from the injured side and minimize pain and discomfort that can further aggravate shock.
- .6.6. avoid rough and excessive handling.
- .6.8. in general, keep an injured patient supine.

Remember, however, that some patients shocked after a severe heart attack or with lung disease cannot breathe as well as when supine as when sitting up or in a semi-setting position. With such a patient, use the most comfortable position and accurately record the victim's pulse, blood pressure, and other vital signs. Maintain a record at 10 minutes interval until the patient is under medical care. *Do not give the victim anything to eat or drink.*

11.3. Hypovolemic Shock (Hemorrhagic shock)

Following injury, shock is commonly a result of fluid or blood loss. It also results from severe thermal burns.

- .3.1. Factor that contribute to continues bleeding
 - .1.1. failure to apply sufficient pressure to obvious external bleeding points.
 - .1.2. failure to splint fracture properly
 - .1.3. failure to handle injuries gently
- .3.2. Causes
 - .2.1. external bleeding
 - .2.2. internal bleeding (follow rupture of liver or spleen)
 - .2.3. injury of blood vessel within the abdomen or chest
 - .2.4. severe thermal burn
 - .2.5. crushing injuries
- .3.3. First Aid (Emergency Support)
 - .3.1. proper position
 - .3.2. ventilatory support
 - 3.3. transport immediately to near emergency department for definitive care.

11.4. Psychogenic Shock

or Fainting called syncope is a sudden reaction of the Nervous System that produce partial or temporary vascular dilation. The result is a temporary, reduction of blood supply to the brain because the blood momentarily pools in the dilated vessel in the other parts of the body.-

- .4.1. Causes
 - .1.1. fright
 - .1.2. sudden news (either good and bad)
 - .1.3. sight of blood
 - .1.4. injury
 - .1.5. death.
 - .1.6. prolonged standby in one spot

- .1.7. witness a horrible accident
- .1.8. fear
- .1.9. anxiety
- .4.2. Indication of Psychogenic shock
 - 2.1. sudden change of behavior
 - 2.2. strange loss of memory
 - 2.3. delusion of grandeur
 - 2.4. nauseous
 - 2.5. feel lightened
 - 2.6. face pale
 - 2.7. tingling or numbness in the extremities
- .4.3. First Aid (Emergency Care)
 - 3.1. elevation of lower extremities
 - 3.2. application of cold compress
 - 3.3. onlookers must be kept distance
 - 3.4. transport victim to emergency department

NOTE: Before transporting the victim try to learn from bystanders how long the victim had been unconscious.

11.5. Neurogenic Shock

Shock that accompanies spinal cord injury is best treated by a combination of all known supportive measures.

- .5.1. Causes
 - .1.1. spinal cord injury
 - .1.2. upper cervical
 - .1.3. injury to the part of nervous system
 - .1.4. perfusion of organs and tissue
- .5.2. First Aid
 - .2.2. proper position
 - .2.3. Basic Life Support is needed
 - .2.4. victim must be kept warm
 - .2.5. prompt transfer to hospital is mandatory

11.6. Metabolic Shock

Metabolic shock is usually the result of an illness that has been present for a long time or has been extremely over a brief period.

- .6.1. Causes
 - .1.1. Diarrhea
 - .1.2. excessive urination
 - .1.3. severe disturbance of body fluid and (uncontrolled disease such diabetes mellitus)
 - .1.4. severely dehydrated
- .6.2. First Aid (Emergency Care)
 - .2.1. transport victim to near hospital
 - .2.2. give all needed support (including oxygen)

11.7. Respiratory Shock (nonvascular causes)

The proper emergency management of shock as a result of inadequate respiration involves the immediate securing and maintaining of an airway.

- .7.1. Cause
 - Obstruction (from the throat down to the larynx (mucus, vomitus and foreign materials)

.7.2. First Aid (Emergency Care)

.2.1. Basic Life Support

.2.2. transport immediately to emergency department

11.8. Septic Shock

In some patients who have severe bacterial infection, toxins (poison) can be produced by the bacteria or by infected body tissue.-

.8.1. Causes

.1.1. damaged or injured vessel walls

.1.2. dilation of vessels

.1.3. loss of plasma

.8.2. First Aid (Emergency Care)

.2.1. elevation of the lower extremities

.2.2. transport immediately to the Hospital

.2.3. respiratory support (oxygen)

NOTE: This type of shock is a complex problem that can lead to a leak of blood in the vascular system (hypovolemia). At the same time, there is a larger than normal blood vessel in an attempt to contain the smaller than normal volume of intravascular blood.

MODULE 5. BASIC LIFE SUPPORT (CARDIOPULMONARY RESUSCITATION): INTRODUCTION AND ARTIFICIAL RESPIRATION BACKGROUND AND GENERAL PRINCIPLES

1. Breathing and Circulation

- 1.1. Air that enter the lungs contains about 21 percent of oxygen and only a trace of carbon dioxide. Air that is exhaled from the lungs contains about 16 percent oxygen and 4 percent carbon dioxide.
- 1.2. The right side of the heart pumps blood to the lungs, where blood picks up oxygen and releases carbon dioxide.
- 1.3. The oxygenated blood then returns to the left side of the heart, where it is pumped to the tissues of the body.
- 1.4. In the body tissue, the blood releases oxygen and takes up carbon dioxide after which it flows back to the right side of the heart.
- 1.5. All body tissues require oxygen, but the brain requires more than any other tissue.
- 1.6. When breathing and circulation stop, this is called *clinical death* (0-4 min.: brain damage not likely; 4-6 min. damage probable).
- 1.7. When the brain has been deprived of oxygenated blood for a period of 6 minutes or more, an irreversible damage probably occurred, this is called *biological death* (6-10 min.: brain damage probable; over 10 minutes brain damage is certain).
- 1.8. It is obvious from the above stated facts that both respiration and circulation are required to maintain life.
- 1.9. When breathing stops, the pulse and circulation may continue for sometime, a condition known as respiratory arrest. In this case only artificial respiration is required since the heart action continues to circulate blood to the brain and the rest of the body.
- 1.10. When circulation stops, the pulse disappears and breathing stops at the same time or soon thereafter. This is called *cardiac arrest*. When cardiac arrest occurs, both artificial respiration and artificial circulation are required to oxygenate the blood and circulate it to the brain.-

2. Cardiac Arrest

At one time the term cardiac arrest indicate that the heart has stopped beating, but it now has a much broader meaning. Cardiac arrest is any of the three conditions describe below in which the circulation is either absent or inadequate to sustain life.

- 2.1. In *cardio vascular collapse* the heart is still beating but its action is so weak that blood is not being circulated through the vascular system to the brain body tissues. This condition may result from hemorrhage or various drugs.
- 2.2. When *ventricular fibrillation* occurs, the individual fascicles of the heart beat independently rather than the usual coordinated, synchronized manner that produce rhythmic heartbeat. Direct inspection of the heart condition reveals an organ that looks and feel like a bag of worms. Ventricular fibrillation sometimes occurs following heart attacks, and it is seen frequently following voltage electric shocks.
- 2.3. *Cardiac standstill* means that the heart has stopped beating. This condition may be terminal and is usually due to lack of oxygen (anoxia) of the heart muscle.

It is important to know that there are various types of cardiac arrest. In an emergency, however, it is not necessary to determine which type of cardiac arrest is present. All three types can be recognized by absent respiration and absent pulse in an unconscious person with a deathlike appearance.

Begin cardiopulmonary resuscitation (CPR) immediately when you recognize cardiac arrest.

3. Life Support

Life support is obviously the goal of cardiopulmonary resuscitation.

Stages of life support are as follows:

- 3.1. *Basic Life Support* - an emergency procedure that consist of recognizing respiratory or cardiac arrest or both and the proper application of CPR to maintain life until a victim recovers or advance life support becomes available.
 - .1.1. Basic A B C steps
 - Airway opened
 - Breathing restored
 - Circulation restored
 - .1.2. Use of supplementary techniques
- 3.2. *Advanced Cardiac Life Support (ACLS)*
 - .2.1. Definitive therapy
 - o Diagnosis
 - o Drugs
 - o Defibrillation
 - .2.2. Cardiac monitoring stabilization
 - .2.3. Transportation
 - .2.4. Communication
- 3.3. *Prolonged Life Support (PLS)* for post resuscitative and long term resuscitation.

CARDIOVASCULAR DISEASE

I. Risk Factors for Cardiovascular Disease

- 1.1. Risk factors that cannot be changed
 - .1.1. heredity
 - .1.2. age
 - .1.3. sex
- 1.2. Risk factors that can be changed
 - .2.1. cigarette smoking
 - .2.2. high cholesterol diet
 - 2.3. high blood pressure
- 1.3. Contributing risk factors that can be changed or controlled
 - .3.1. obesity
 - .3.2. lack of exercise
 - .3.3. diabetes

2. Heart Attack (Myocardial Infraction)

A heart attack occurs when the oxygen supply to the heart muscle (myocardium) is cut off for a prolonged period of time. This cut-off result from a reduced blood supply due to severe narrowing or complete blockage of the diseased artery. The result is death (infraction) of the affected part of the heart.

2.1. Warning signals

- .1.1. chest discomfort or pain
- .1.2. uncomfortable pressure, squeezing, fullness or tightness, aching, crushing, constricting,oppressive or heavy.
- .1.3. sweating
- .1.4. nausea
- .1.5. shortness of breath

2.2. First Aid

- .2.1. recognize the signals of a heart attack and take action.
- .2.2. have the victim stop what he or she is doing and sit or lie down in a comfortable position. Do not let the victim move around.
- .2.3. have someone call the physician or ambulance for help.
- .2.4. if victim is under medical care, assist him in taking his/her prescribed medicine/s.

RESPIRATORY EMERGENCY AND ARTIFICIAL RESPIRATION

- 1. Respiratory Arrest - when breathing stops and circulation continue for quite sometime.
- 2. Causes of respiratory emergency/arrest
 - I.1. Obstruction
 - .I.1. Anatomical obstruction - when tongue drops back and obstruct the throat. Other causes are acute asthma, croup, diphtheria and swelling.
 - .I.2. Mechanical obstruction - when foreign objects lodge in the pharynx or airways; fluids accumulate in the back of the throat.
 - I.2. Disease
 - I.3. Other causes of respiratory arrest
 - .3.1. electrocution
 - .3.2. circulatory collapse
 - .3.3. external strangulation
 - .3.4. chest compression
 - .3.5. drowning
 - .3.6. poisoning
 - .3.7. suffocation
- 3. ARTIFICIAL RESPIRATION (Rescue Breathing)
 - a procedure for causing air to flow into and out of the lungs of a person when his natural breathing ceases or is inadequate.
- 4. Methods of Artificial Respiration Introduced
 - 4.1. Bouncing method
 - 4.2. Rolling method
 - 4.3. Upside down pulling
 - 4.4. Chinese method
 - 4.5. Shuffer method
 - 4.6. Sylvester method (chest pressure arm-lift method)
 - 4.7. Holger-Nielsen method (back-pressure arm-lift method)
 - 4.8. Rescue breathing - direct blowing of air into the air passages of the victim.
 - Note: Rescue Breathing (mouth-to-mouth/nose/mouth and nose/stoma) is the most effective and practical. Hence, the only method to be adopted.
- 5. Objectives of Artificial Respiration
 - 5.1. To open airway
 - .I.1. maximum head-tilt/chin lift method
 - .I.2. jaw thrust maneuver
 - 5.2. To ventilate the lungs
- 6. Important Aspects of Artificial Respiration
 - 6.1. get started immediately.
 - 6.2. apply artificial respiration 10 to 12 times per minute or 1 breathe of 1.5 to 2 seconds, every 5 seconds (adult).
 - 6.3. maintain normal body temperature as supplementary help.
 - 6.4. continue giving artificial respiration even during transportation, if still needed.
 - 6.5. stabilize the victim for quite sometime after recovery.
- 7. Guidelines in Giving Rescue Breathing (Mouth-to-mouth/nose) following the ABC steps:

Step/Activity	:	Critical Performance	:	Rationale
1. Check for unrespon-siveness	:	Tap or shake gently and shout, "Are you okey?"	:	One concern is the risk of unnecessarily resuscitating sleepers, fainters, etc.
	:		:	
	:		:	
	:		:	
	:		:	

2. Call for help	:	Call for "Help"	:	Call for help will summon nearby bystanders. If someone immediately responds, no. 8 below may be carried out, though no complete information about the victim can be given yet.
3. Position the victim	:	Turn if necessary, support the head and neck. Take adequate time.	:	Frequently, the victim victim will be faced downward. Effective AR/CPR can only be provided with the victim flat on the back . The head cannot be above the level of the heart or CPR is ineffective if to be performed.
4. Open airway	:	Kneel beside the victim's shoulder, upper hand on forehead, lower hand on the bony part of the jaw. Press the forehead downward while lifting the chin so that the teeth are nearly brought together. Avoid completely closing the mouth.	:	Airway must be opened to establish breathlessness. Many victims may be making effort for respiration that are ineffective because of obstruction by the tongue.
5. Establish breathlessness (look, listen, and feel for 3-5 seconds).	:	Turn your head toward victim's legs with your ear directly over and close to the victim's mouth. Listen and feel for evidence of breathing. Look for the rise and fall of the chest.	:	Hearing and feeling are the only true ways of determining the presence of breathing. If there is chest movement but you cannot feel or hear air, the airway is still obstructed.
6. If breathless, give two ventilations at 1.5 to 2 sec. per ventilation	:	Pinch off the nostrils with thumb and forefinger of the upper hand while maintaining pressure on the victim's forehead to keep the head tilted. Open your mouth widely, take a deep breath and make a tight seal.	:	When you begin rescue breathing, it is important to get as much oxygen as possible to the victim. If your rescue breathing is effective, you will

	:	Breathe into the victim's	:	feel air going in
	:	mouth 2 times. Watch	:	as you blow, and
	:	the victim's chest rise.	:	feel the resistance
	:		:	of the lungs.
	:	Feel your lungs	:	
	:	emptying. See the	:	
	:	rise and fall of the	:	
	:	victim's chest and	:	
	:	belly.	:	
	:		:	
	:	Ventilation must be	:	Avoid over or under
	:	given from 1.5 to 2 sec.	:	ventilation. Over
	:	and wait for the full	:	ventilation causes
	:	deflation of the chest	:	stomach distention.
	:	before giving the second	:	
	:	breath.	:	
	:		:	
7. Establish	:	Place 2-3 fingers on the	:	This activity should
pulseless-	:	adam's apple and slide	:	take 5 to 10 seconds
ness for	:	into the groove between	:	because it takes
5 to 10	:	the voice box and muscle	:	time to find the
secs.	:	on the rescuer's side.	:	right place and the
	:	Other hand maintain the	:	pulse itself may be
	:	head tilt. Palpate pulse	:	slow or very weak
	:	for 5 to 10 seconds.	:	and rapid. The vic-
	:	Everytime pulse is	:	tim's condition must
	:	checked, breathing is	:	be properly
	:	also simultaneously	:	assessed.
	:	checked.	:	
	:		:	
8. Activate	:	Know your local medical	:	Notification to the
medical	:	services telephone num-	:	medical services at
assistance	:	ber. Send someone to	:	this time allows the
or transfer	:	call.	:	caller to give
facility.	:		:	complete information
	:		:	about the victim's
	:		:	condition.
	:		:	
	:	In most cases, ask	:	It would be imprac-
	:	someone to arrange for	:	tical to ask some-
	:	transfer facility.	:	body call for med-
	:		:	cal services if
	:		:	there is no tele-
	:		:	phone available or
	:		:	no physician/hospi-
	:		:	tal within the
	:		:	vicinity.
	:		:	
9. If victim's	:	Begin 1 rescue breathing	:	If the heart is
pulse is	:	every 5 seconds. Watch	:	still beating and
present but	:	chest deflate after each	:	circulating blood,
not breath-	:	ventilation. Continue	:	Increasing the
ing. Give	:	rescue breathing for 1	:	oxygen level may
one breathe	:	minute (10 to 12 breaths);	:	stimulate the
every 5 se-	:	check pulse for 5 sec.	:	breathing control
conds.	:	and resume or stop res-	:	center and the vic-
	:	cue breathing as indi-	:	tim may resume to
	:	cated.	:	have normal breath
	:		:	ing.
	:		:	
	:		:	

10. Place victim :	Turn the victim to his	:	Once breathing
in recovery :	side (away from you).	:	is restored vom-
position af- :	Lower arm may be taken	:	iting or regurgita-
ter breathing :	advantage as a pillow.	:	tion may occur
is restored. :		:	anytime.

Note: For standardization purposes, mnemonic of 1 breathe every 5 seconds is as follows: breathe (1.5 - 2 seconds), catch your breathe (.5 sec.)

...one—(.5 sec.)(= 1 sec.);
 ...one thousand—(.5 sec.) *two*—(.5 sec.)(=2 secs.);
 ...one thousand—(.5 sec.) *three*—(.5 sec.)(=3 secs.);
 ...one thousand—(.5 sec.) ONE....(the counting number of breathes) (.5 sec.) (=4 secs.);
 ...take a deep breath (.5 sec.), breathe (this is the 5th second though the breathe is to be given from 1.5 to 2 seconds).

That is the complete cycle of 1 breathe every 5 seconds.

Again: catch breathe _ ONE; one thousand two; one thousand three; one thousand _ TWO—breathe—..... until 10, 11, or 12 (approximately 1 minute).

8. The Modified Jaw Thrust Maneuver

- used to open the airway when the rescuer suspects that the victim has a head, neck, or back injury, because it minimizes head and neck movement.

A head, neck, or back (spinal cord) injury should always be suspected in victims who have been in a violent accident or who have suffered a traumatic injury, particularly if the trauma might have subjected the spine to sudden acceleration or deceleration. This could be from a vehicular accident, fall, diving accident or other sports-related accident. If there is a head injury and the victim is unconscious, the rescuer should suspect a spinal cord injury. If a spinal cord injury is suspected, the rescuer immediately kneels behind the victim and stabilizes the the victim’s head and neck (keeps the head still). The rescuer places his/her hands along both sides of the victim’s head with the fingers touching the jaw line prevent the head from moving from side to side to forward and backward. This technique is known as the “in_line stabilization” because it keeps the head in line with the spine. Then during the primary survey, when checking for unresponsiveness in a victim who may have head, neck or back injury, the rescuer asks, rather than shouts, “Are you OK?”. This is done so the the victim is not startled, which might cause him/her to move or jerk in surprise, causing further injury. If a head, neck, or back injury is suspected, the head should not be turned to the side or the body moved. If moving the victim is necessary to deliver basic life support, the head, neck and back should supported and turned as a unit. It is recommended that more than one person help turn the victim, working together so the victim rolled as a one unit. The modified jaw thrust maneuver should then be used to open the airway. To perform the modified jaw thrust, the rescuer kneels at an angle behind the victim’s head, positions hi/her elbows on the surface on which the victim is lying, and rests his/her hands on both sides of the victim’s head to support it and keep it immobile. The rescuer places the fingers of both hands under the victim’s lower jaw just in front of the earlobes, positions the thumbs across the victim’s cheekbones, and then applies pressure upward to lift the jaw forward and open the airway. The rescuer then performs rescue breathing as described in preceding pages.9. Mouth_to_Nose Rescue Breathing There are a few situations when the rescuer may not be able to make a tight enough seal over a victim’s mouth to perform mouth_to_mouth rescue breathing. For example, the victim’s jaw or mouth may be injured during an accident, the jaw may be shut -_H_’- 5_9_â+h) 0*0*0*_+î too tight to open, or the rescuer’s mouth may be too small. In such cases, mouth_to_nose rescue breathing should be done as follows: 9.1. The rescuer maintains the backward head_tilt position with one hand on the victim’s forehead, and uses the other hand to close the mouth, being sure to push on the chin and not on the throat. 9.2. The rescuer open his/her mouth wide, takes a deep breath, seals his/her mouth tightly around the victim’s nose and breathes full breaths into the nose, doing the skill as described for the mouth_to_mouth

